Empirical Validation Plan

HIGH Priority Validations

V1. Collection Validation

Goal: Measure whether the collectors retrieve all relevant papers.

Steps:

1. Search for SMSs (preferably in the same field) that shared the papers dataset they used.

   $\Rightarrow$ The selected datasets will be called *standard-datasets*.

2. Select about 10 SMSs.

3. Run the collection with the same search strings and year range.

4. Compute:

   - Recall = $\frac{\text{papers from standard-dataset found by the tool}}{\text{total standard-dataset papers}}$
   - Per-source recall = breakdown recall by ACM, IEEE, Springer, Scopus.

V2. Deduplication Precision/Recall

Goal: Validate that the two-pass deduplication removes ALL duplicates AND ONLY duplicates.

Steps:

1. Manually label a set of 500 to 1000 papers as duplicate or not-duplicate.

2. Run the deduplication on the same set.

3. Compute:

   - Precision = $\frac{\text{true duplicates caught}}{\text{all flagged as duplicate by the tool}}$
   - Recall = $\frac{\text{true duplicates caught}}{\text{all actual duplicates}}$
   - F1 score

4. Sensitivity analysis: Vary the TF-IDF threshold (0.80, 0.85, 0.90, 0.92, 0.95) and plot precision-recall curve to choose the best value.

5. Analyze false positives (non-duplicates deleted by the script) and false negatives (duplicates not detected by the script).

V3. Automated Exclusion Accuracy

Goal: Validate the LLM-based automated exclusion.
Steps:

1. Take 200 to 500 papers and label them with relevance (`KEEP`/`REMOVE`).
2. Run automated exclusion on the same papers.
3. Compute:
   - Precision = $\frac{\text{correctly kept}}{\text{all kept by LLM}}$
   - Recall = $\frac{\text{correctly kept}}{\text{all that should've been kept}}$
   - Specificity = $\frac{\text{correctly removed}}{\text{all that should've been removed}}$
   - Cohen's Kappa (inter-rater agreement between LLM and human)
4. Analyze the error types:
   - False exclusions (relevant paper removed) — *most dangerous*
   - False inclusions (irrelevant paper kept) — less dangerous (will be removed in screening anyway)

V4. Dual-LLM Screening Agreement & Accuracy

Goal: Validate the SMS-screening dual-model conflict resolution.
Steps:

1. Prepare a stratified sample of 100–200 papers across relevance levels.

   $\Rightarrow$ i.e., divide 100–200 papers by relevance level and select some papers from each group, instead of taking papers randomly from the whole dataset.
2. Independently (to have 2 human reviews) screen each paper, then define the ground truth.
3. Run the screening on the same papers.
4. Compute:
   - Per-model accuracy (Cerebras vs. Groq separately against human ground truth)
   - Inter-model agreement (Cohen's Kappa between Cerebras and Groq)
   - Conflict resolution accuracy: "Does the confidence-based resolution choose the correct answer?"
   - Human-vs-system Kappa (final system decision vs. human consensus)
5. Disagreement analysis: "When do models disagree?", "Which is right more often?"
6. Evaluate the manual review flag: "Of papers flagged, what % actually needed correction?"

V5. Screening Reproducibility

Goal: Measure LLM screening stability across repeated runs.

Steps:

1. Screen the same 100 papers 3 times at different times (different API calls).
2. For each paper, record decision + confidence across all runs.
3. Compute:
   - Decision flip rate = $\frac{\text{papers that changed decision across runs}}{\text{total}}$
   - Intra-class correlation (ICC) on confidence scores
   - Fleiss' Kappa across 3 runs (treating each run as a "rater")
4. Stratify by confidence level: "Are low-confidence papers (< 0.7) the ones that flip?"

V6. Snowballing Coverage Validation

Goal: Validate that forward/backward snowballing discovers relevant papers not in the original search.

Steps:

1. From a completed screening, take all `INCLUDE` papers with snowballing results.
2. Ground truth: Manually extract references from 20 to 30 papers, manually search Google Scholar for citations.
3. Compute:
   - Backward recall = $\frac{\text{S2 references found}}{\text{manually extracted references}}$
   - Forward recall = $\frac{\text{S2 citations found}}{\text{Google Scholar citations}}$
   - New discovery rate = $\frac{\text{snowballed papers not in original collection}}{\text{total snowballed}}$
   - Relevance rate of discoveries = $\frac{\text{relevant snowballed papers}}{\text{total snowballed (after LLM filtering)}}$
4. Compare S2 API coverage vs. OpenAlex fallback vs. text-parsing fallback.

V7. Data Extraction Quality (Paper Extraction Agent)

Goal: Validate LLM-extracted structured data against human annotation.

Steps:

1. Manually extract the same fields from 30 to 50 papers:
   - Metadata (title, authors, year, venue, DOI)
   - Method type / approach description
   - Targeted vulnerabilities
   - Datasets used
   - Key results (precision, recall, F1)
   - Limitations

2. Compare LLM extraction vs. human extraction field-by-field.
3. Compute:
    - Metadata accuracy = exact match rate for title, year, venue, DOI
    - Method type accuracy = correct classification rate
    - Factual accuracy = for numeric fields (precision/recall/F1), compute mean absolute error
    - Completeness = $\frac{\text{fields filled by LLM}}{\text{fields filled by human}}$
    - Hallucination rate = fields where LLM states something NOT in the paper
    - ROUGE/BERTScore for free-text fields (approach description, limitations, etc.)

V8. Method Categorization Agreement

Goal: Validate LLM-driven method categorization against expert taxonomy.
Steps:

1. Independently categorize 50–100 papers by method type.
2. Run LLM categorization on the same papers.
3. Compute:
    - Agreement with experts (Cohen's Kappa for each person–LLM pair)
    - Granularity appropriateness: "Are categories too broad or too specific?"
    - Exclusive assignment correctness: "For hybrid papers, does the LLM pick the right primary method?"
    - Category stability — re-run categorization 3 times, measure consistency.
4. Compare LLM taxonomy against established taxonomies (e.g., CWE top-25, OWASP classification).

V9. End-to-End Comparison with Manual SMS

Goal: Compare the full automated pipeline against a manually conducted SMS.
Steps:

1. Baseline: Take a published, peer-reviewed SMS in the software security domain.
2. Run the full pipeline on the same research questions, search strings, and year range.
3. Compare at every stage:

Stage	Metric	What to Compare
Collection	Paper set overlap	Jaccard similarity of paper sets
Deduplication	Duplicate detection rate	Same duplicates identified?
Screening	Inclusion set overlap	Papers INCLUDE'd by both
Screening	Exclusion agreement	Papers EXCLUDE'd by both
Snowballing	New papers discovered	Unique papers found by each method
Extraction	Data quality	Side-by-side field comparison
Categorization	Taxonomy alignment	Same groupings?

4. Compute:
    - Overall pipeline recall = $\frac{\text{manual SMS papers found by automated}}{\text{total manual SMS papers}}$
    - Overall pipeline precision = $\frac{\text{relevant papers in automated set}}{\text{total automated set}}$
    - Time comparison = hours spent manual vs. automated
    - Cost comparison = API costs vs. researcher hours

Less Priority Validations

V10. Sensitivity Analysis of Key Parameters

Goal: Understand how configuration choices affect outcomes.
Parameters to vary:

Parameter	Range to Test	Affected Stage
TF-IDF similarity threshold	0.80 – 0.98 (steps of 0.02)	Deduplication
LLM temperature	0.0, 0.1, 0.2, 0.3, 0.5	Screening + Extraction
Confidence threshold	0.5, 0.6, 0.7, 0.8, 0.9	Conflict resolution
Text truncation limit	10K, 20K, 30K, 50K chars	Extraction quality
Section cap	1000, 2500, 5000 chars	Screening prompt quality
Jaccard PDF title match	0.7, 0.8, 0.85, 0.9	PDF finder accuracy
Min publication year	2000, 2005, 2010, 2015, 2020	—
Relevance filter batch size	10, 30, 50	Snowballing filter quality

For each: Plot the effect on downstream accuracy (screening F1, extraction completeness, etc.)

V11. Model Comparison

Goal: Justify the choice of the two LLMs.
Steps:

1. Take 100 papers with human ground-truth screening decisions.
2. Run screening with multiple models.
3. Compare: accuracy, cost per paper, latency, agreement rate.
4. Test truly independent models for dual-screening (e.g., Llama vs. GPT-4o) to measure whether architectural diversity improves conflict resolution accuracy.

V12. Scalability & Performance Benchmarking

Goal: Measure system behavior under realistic loads.
Steps:

1. Measure wall-clock time per stage for corpora of 100, 500, 1000, 5000 papers.
2. Track:
    - Collection throughput (papers/minute per scraper)

- Deduplication time complexity (should be $O(n^2)$ for TF-IDF)
- Screening throughput (papers/minute, limited by LLM API rate)
- PDF retrieval throughput (limited by PDF source rate limits)
- Extraction throughput
- Identify bottlenecks (likely: LLM API rate limits, PDF downloads)
- Measure resource usage: memory, CPU, network bandwidth

V13. Error Propagation Analysis (Cascade Failures)

Goal: Measure how errors in early stages compound downstream.
Steps:

1. Inject known errors at each stage:
   - Collection: Remove 10%, 20%, 30% of papers randomly
   - Deduplication: Deliberately merge 5% of non-duplicate papers
   - Screening: Randomly flip 5%, 10%, 15% of decisions
   - PDF retrieval: Simulate 20%, 40% retrieval failure
2. Measure the final output quality (extraction completeness, method categorization accuracy, solution relevance) after each injection.
3. Plot error amplification curves: "Does a 10% screening error cause 10% extraction error, or more?"

V14. User Study (Usability & Trust)

Goal: Validate that the tool is practical for real researchers.
Steps:

1. Have 5 to 10 people in the field.
2. Each participant conducts a mini-SMS using our tool (e.g., 50 papers, specific subtopic).
3. Measure:
   - Task completion rate: "Can they finish the full pipeline?"
   - Time-on-task: "How long for each stage?"
   - Error rate: "How often do they need to intervene?"
   - Trust calibration: "Do they trust the LLM decisions?" (Likert scale)
   - SUS score: System Usability Scale questionnaire
   - Override rate: "How many LLM decisions do they change manually?"
   - Post-study interview: "What worked? What didn't? What's missing?"